

Ideas

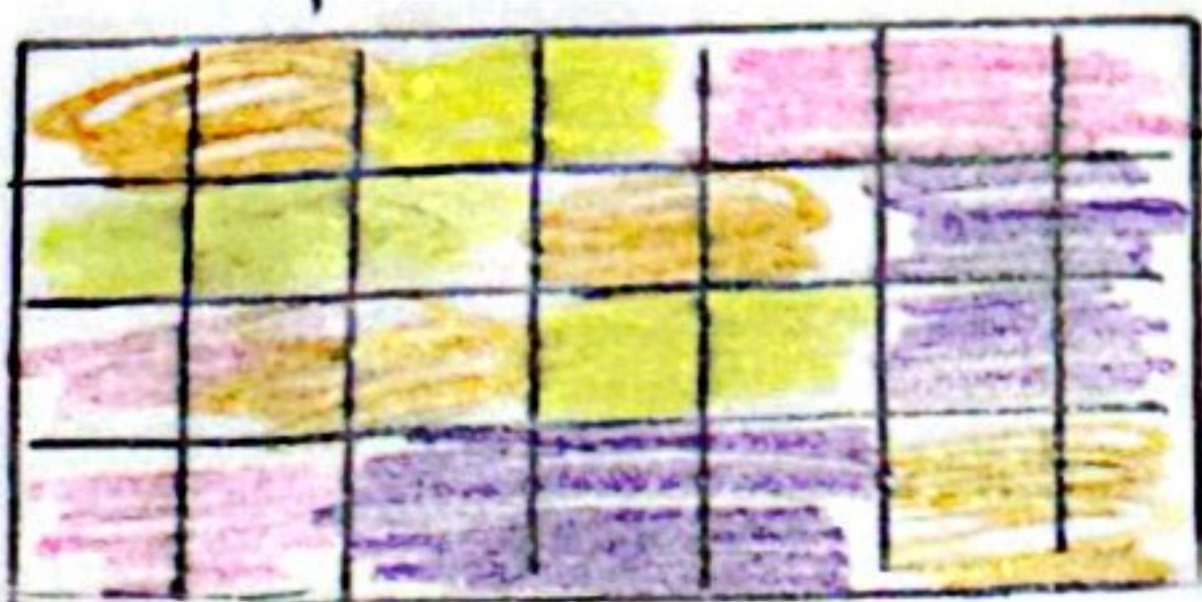
① Choropleth map



total emission per state (2017-2022)

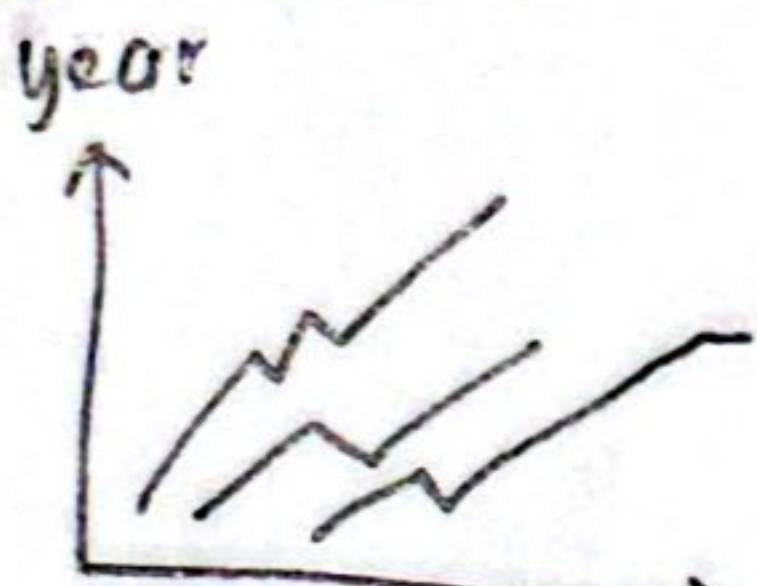
- map diff view - clr shows increase/decrease since 2017

② heatmap



emission changes by state & year

③ line chart



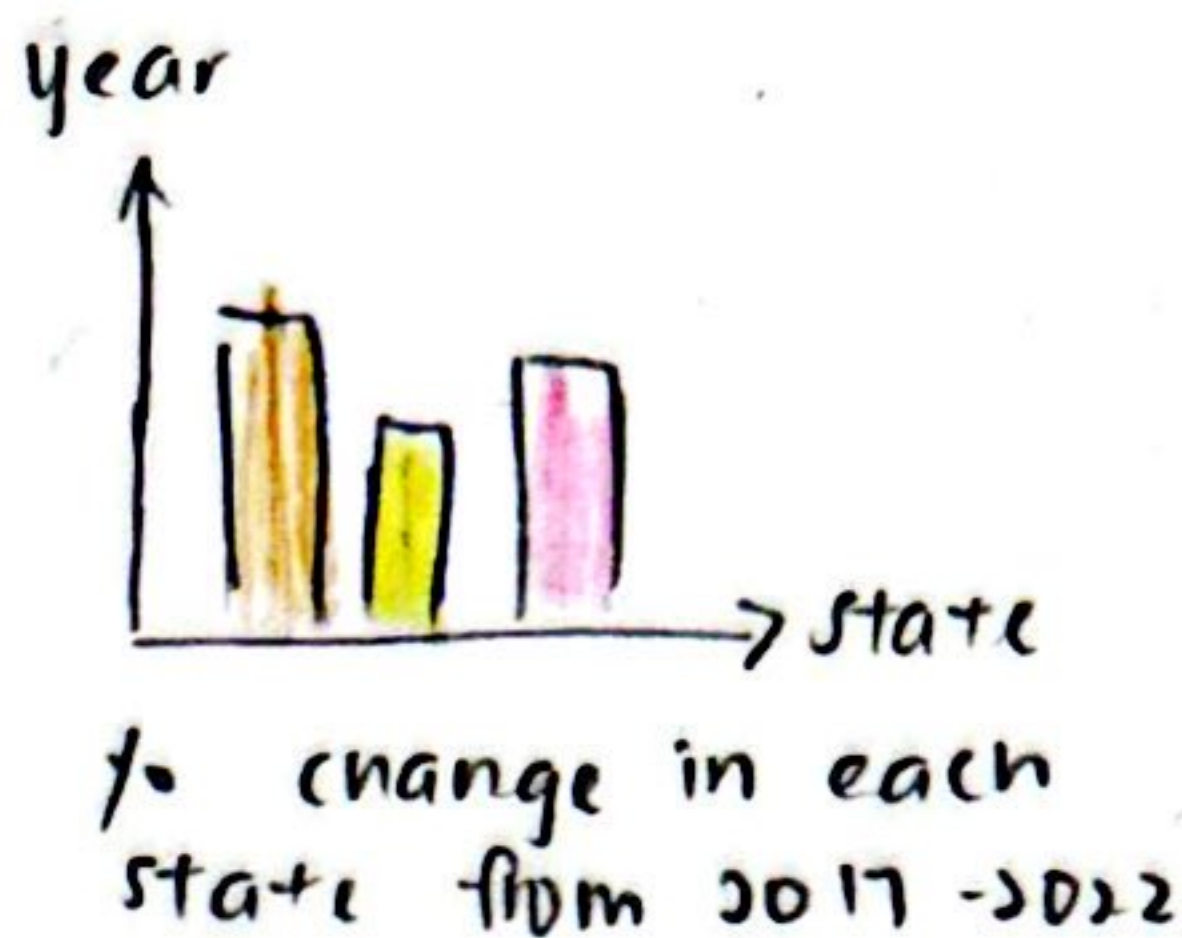
multiple small multiple - 1 line/state

④ stack area chart

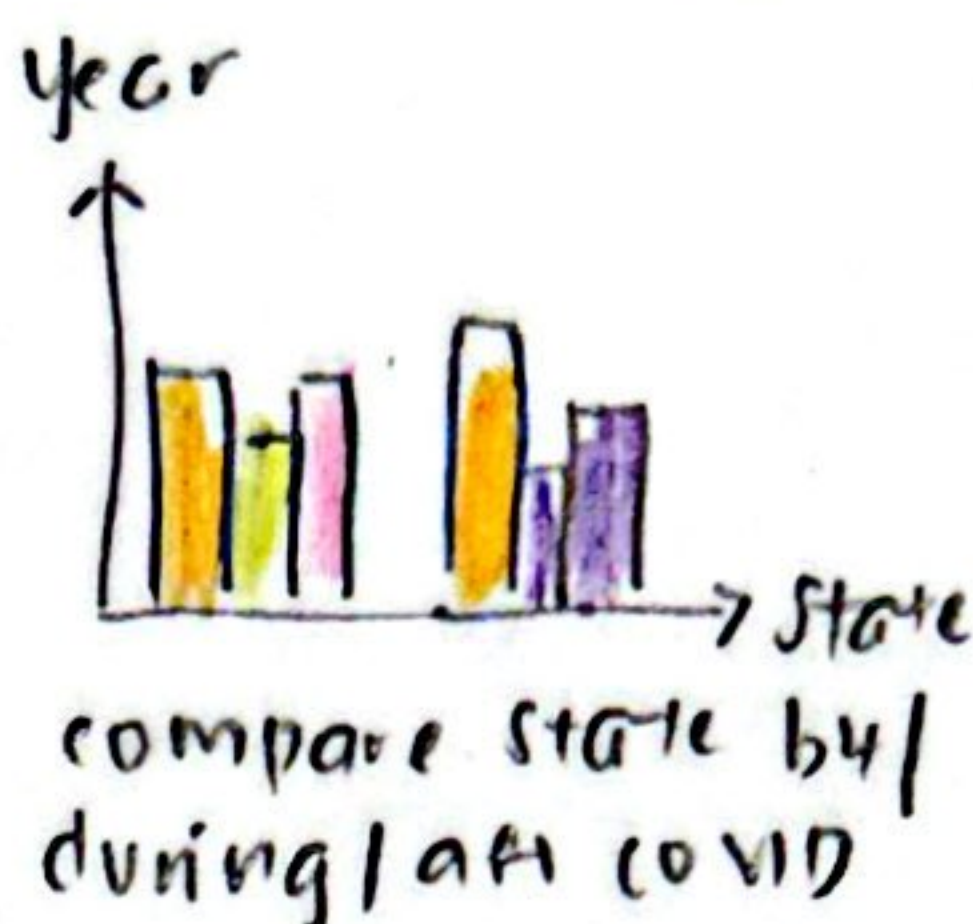


national by sector (energy, transport, waste etc)

⑤ Bar chart

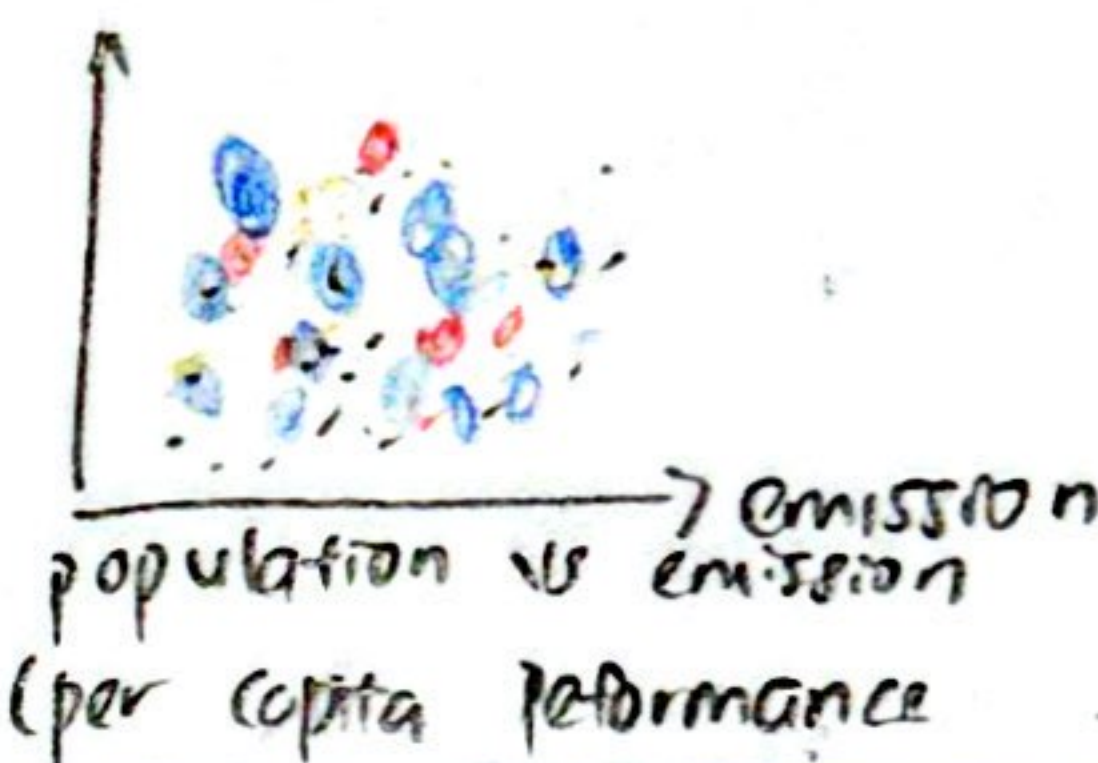


⑥ grouped bar chart



emission intensity vs GDP (economic efficiency)

⑦ scatter plot



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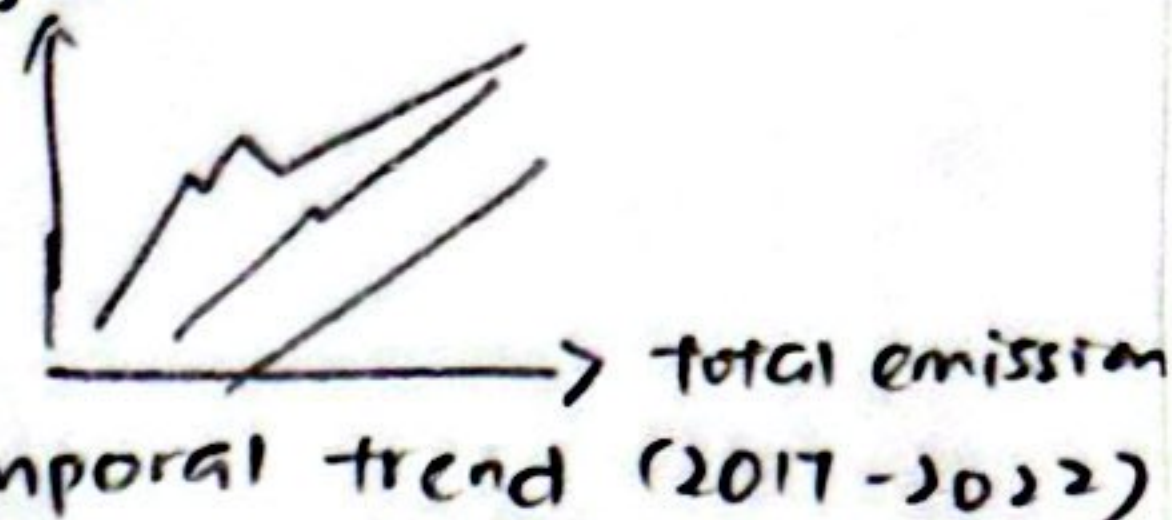
Filter

① map

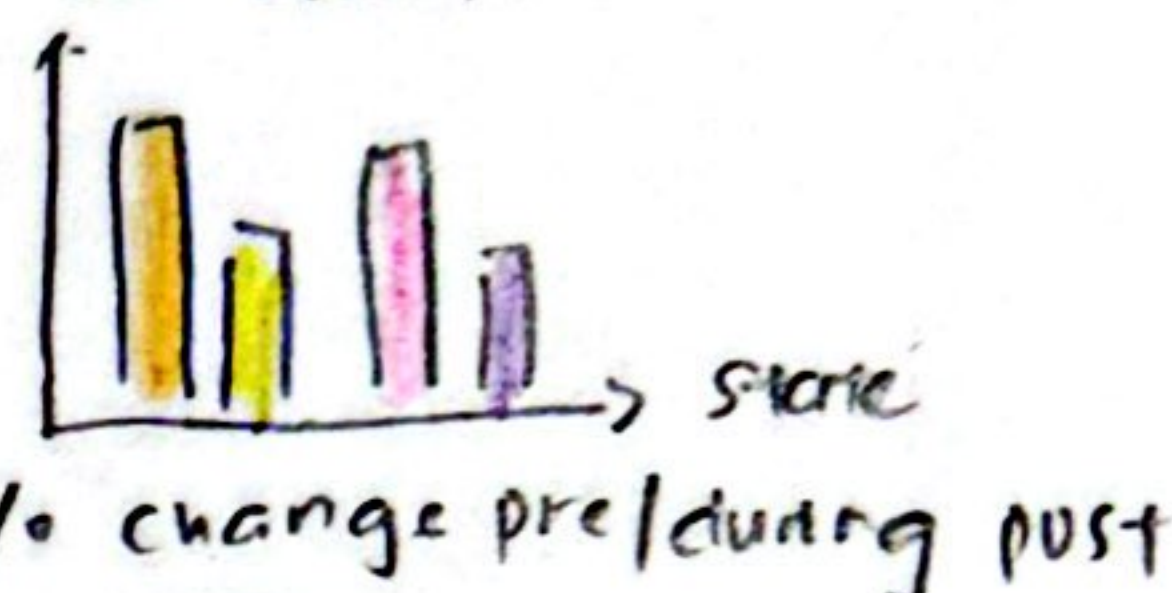


choropleth + interactive state filter

② line chart

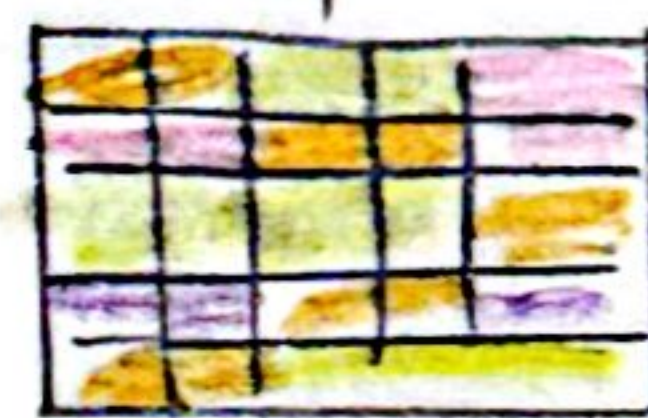


③ Bar chart



emission intensity vs GDP

④ heatmap



emission by state & year

Combine & refine

① Spatial + temporal rs

- choropleth - total emission / state
- line - emission trend

→ where & where emission changed
→ click state on map, line chart update
→ focus geographical context & temporal behavior

② Comparative + analysis

- bar - compare % by, during, after covid
 - bubble - relates emission intensity
- highlight performance & efficiency diff state

③ Trend overview / pattern summary

- heatmap - colour coded change by state & year

→ provide compact nationwide summary of emission lvl & pattern thru covid period

Category

① Geographic + Map (choropleth + interactive state filter)

- total greenhouse gas emission / state
- click state to filter all linked chart

② temporal - line (temporal trend 2017-2022)

- how state's total emission changed by, during after covid

③ Comparative - bar (% change by, during, after)

- compares overall emission changes between diff state

④ Analytical - bubble (emission intensity vs GDP)

- relate each state's economic output to its emission intensity

⑤ Pattern / Trend overview - heatmap (emission change by state & year)

- uses colour to show lvl / % changes for all state

Question

① Does this explain how COVID-19 affected AUS emission 2017-2022?

- ① " show diff between states clearly?
- ② " capture trend of drop & rebound effectively?
- ③ " reveal rs between emission intensity & GDP
- ④ " reveal rs between emission intensity & GDP
- ⑤ Is it too crowded / tell clear & coherent story?

Layout

Total GHG emission by state



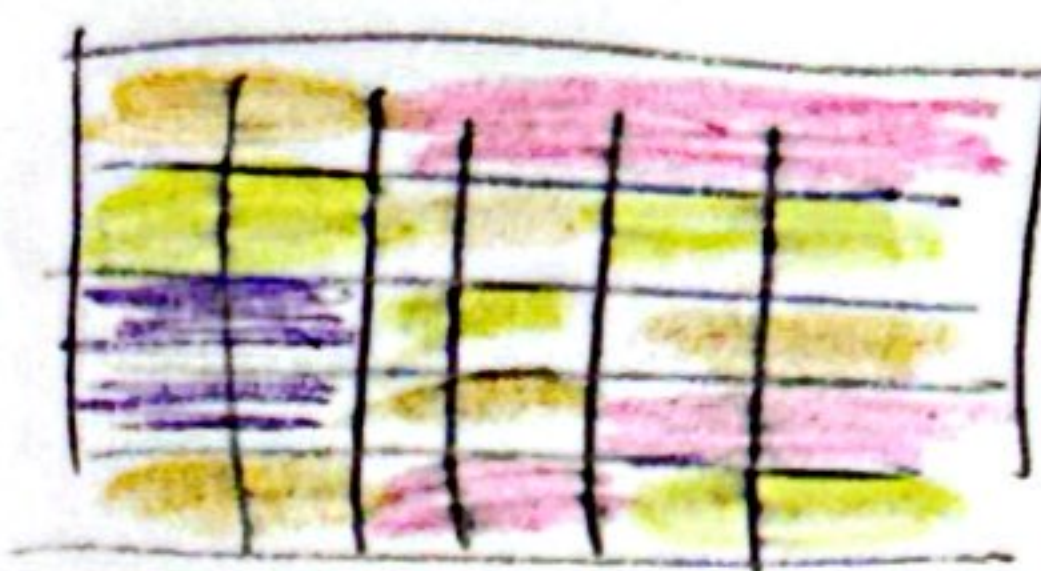
Temporal trend Emission



National emission by sector



Emission intensity



Show economic efficiency VS emission trend



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Operation

- ① click state on map
→ filter line chart & heatmap
- ② hover on line
→ show exact value
+ % change
- ③ drag yr slider
→ map & area chart update for that yr
- ④ hover on heatmap cell
→ show emission intensity
- ⑤ click reset
→ returns to national overview

Discussion

Pros

- covers geographic, temporal, sectoral & economical dimensions in one dashboard
- linked interactivity helps users explore trends & patterns
- supports policy comparison & public understanding
- visually balanced layout makes data story clear & engaging

Cons

- 5 chart increase complexity; req good layout & clr coordination
- too many interactive elements may confuse non-expert users
- performance could slow with large dataset
- need consistent units & scales across charts
→ avoid misinterpretation

Focus

① Key technique

- uses coordinate multiple views (CMV)
- interactions on one chart update all others thru shared filtering (map → line, heatmap, bubble)

② Novel visualization focus

- connects emissions with economic & intensity metrics, letting user see efficiency trends, not just totals.

③ Special detail

- includes a 2020 annotation for COVID-dip, colour-blind friendly diverging palette & dynamic tooltip showing % change since 2017.

④ combination effect

- five linked chart (map, line, stacked area, heatmap, bubble) work tgt to reveal relationships between location, time & performance, giving users both overview detail at once.

Layout

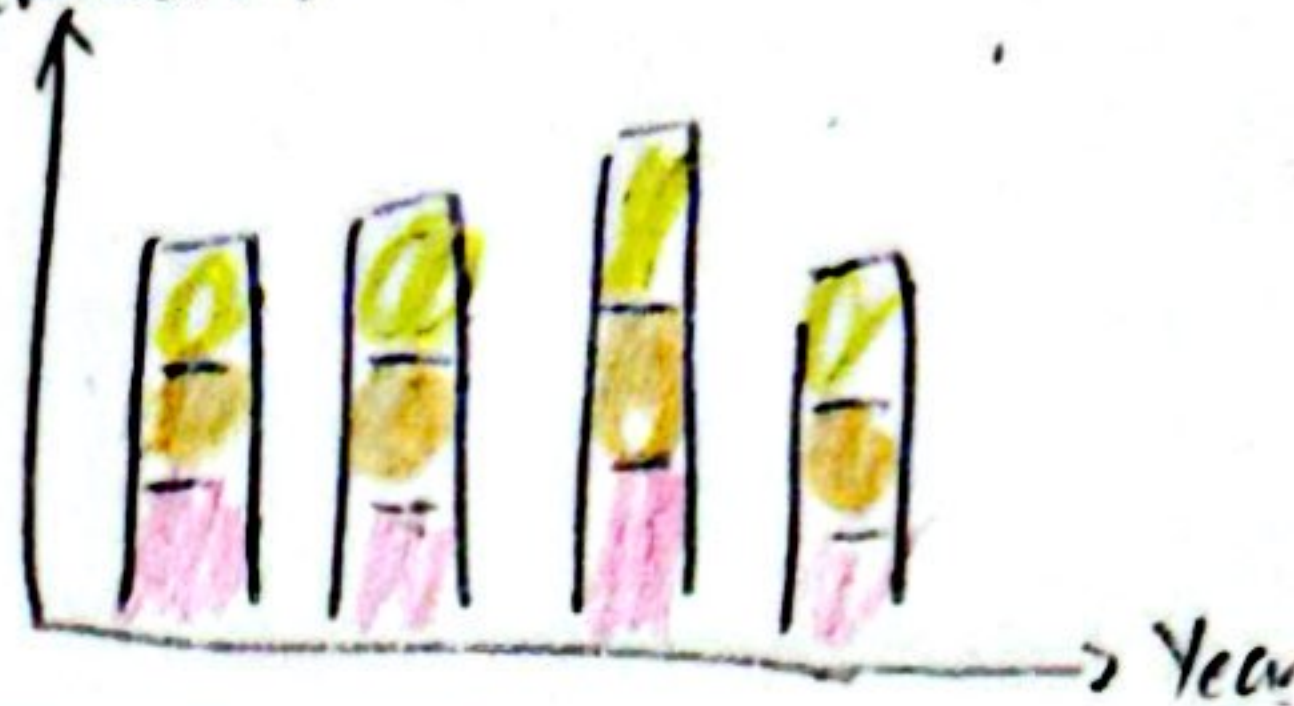
Total emission
by state



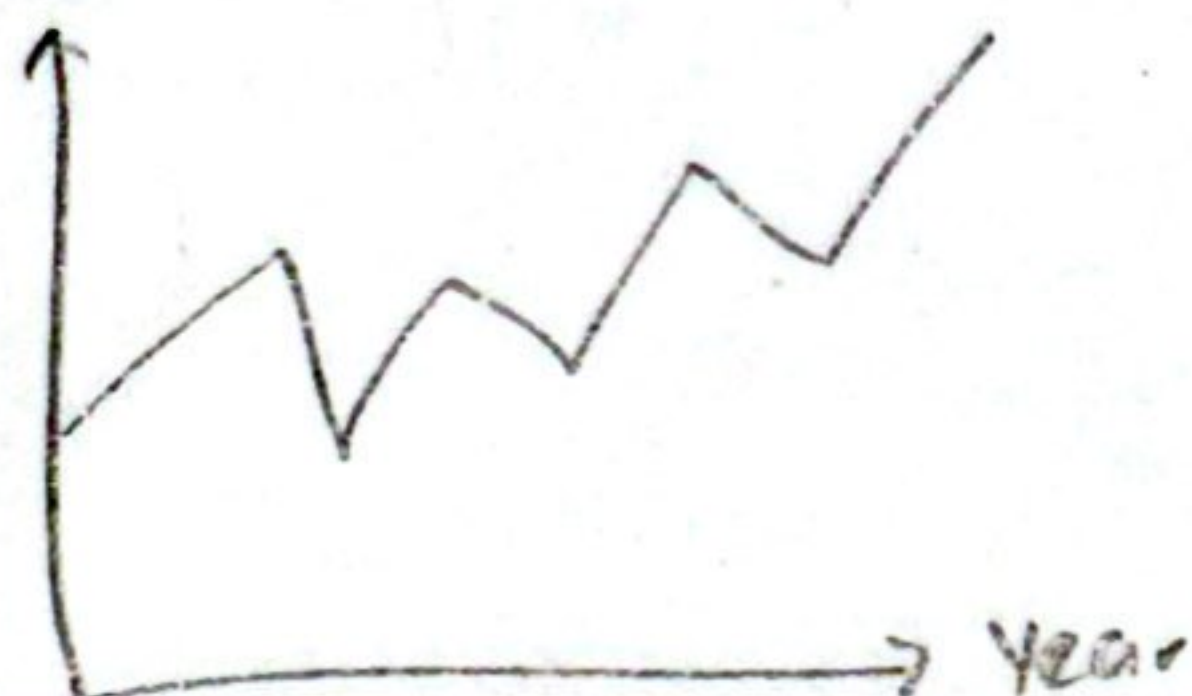
National trend



Compare emission of all
sector
emission



Emission intensity
Emission



Sector share for selected
state & year



Focus

① Key technique

- uses coordinated multiple views (CMV)
- selecting a sector / state updates all charts simultaneously

② Novel visualization focus

- combines choropleth + stacked area + donut + line
- geography, time, & composition in one view

③ Special detail

- sector colour palette stays consistent across all charts, map shading transition smoothly when year slider moves

④ combination effect

- 5 linked charts provide a balanced picture of where emission occur, which sectors dominate & how they change through time

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Operation

① select state on map
→ update bar, donut, line charts

② select sector
→ recolor map + filter all charts

③ hover
→ tooltip with emission value & % change

④ move year slider
→ animate map & stacked area chart

Discussion

Pros

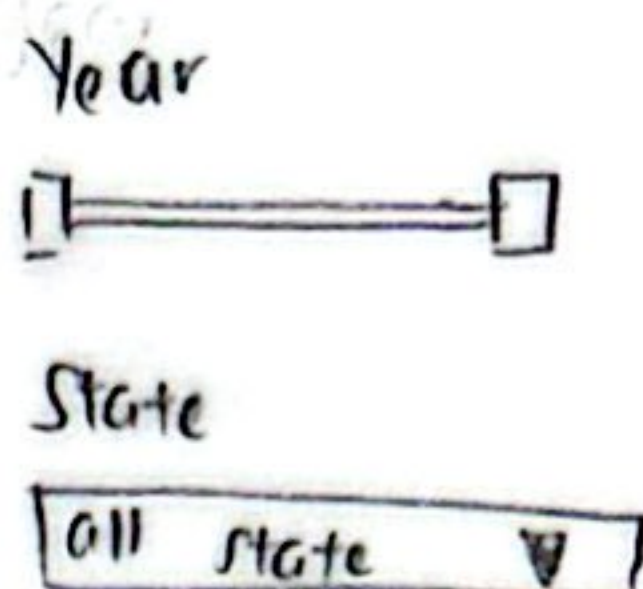
- connect spatial, sectoral, temporal data in one interface
- consistent colour scheme enhance readability
- easy to see which sectors dominate each state

Cons

- multiple chart types need careful spacing
- too many sectors may make bar chart crowded
- interactive filters may require strong guidance labels

Layout

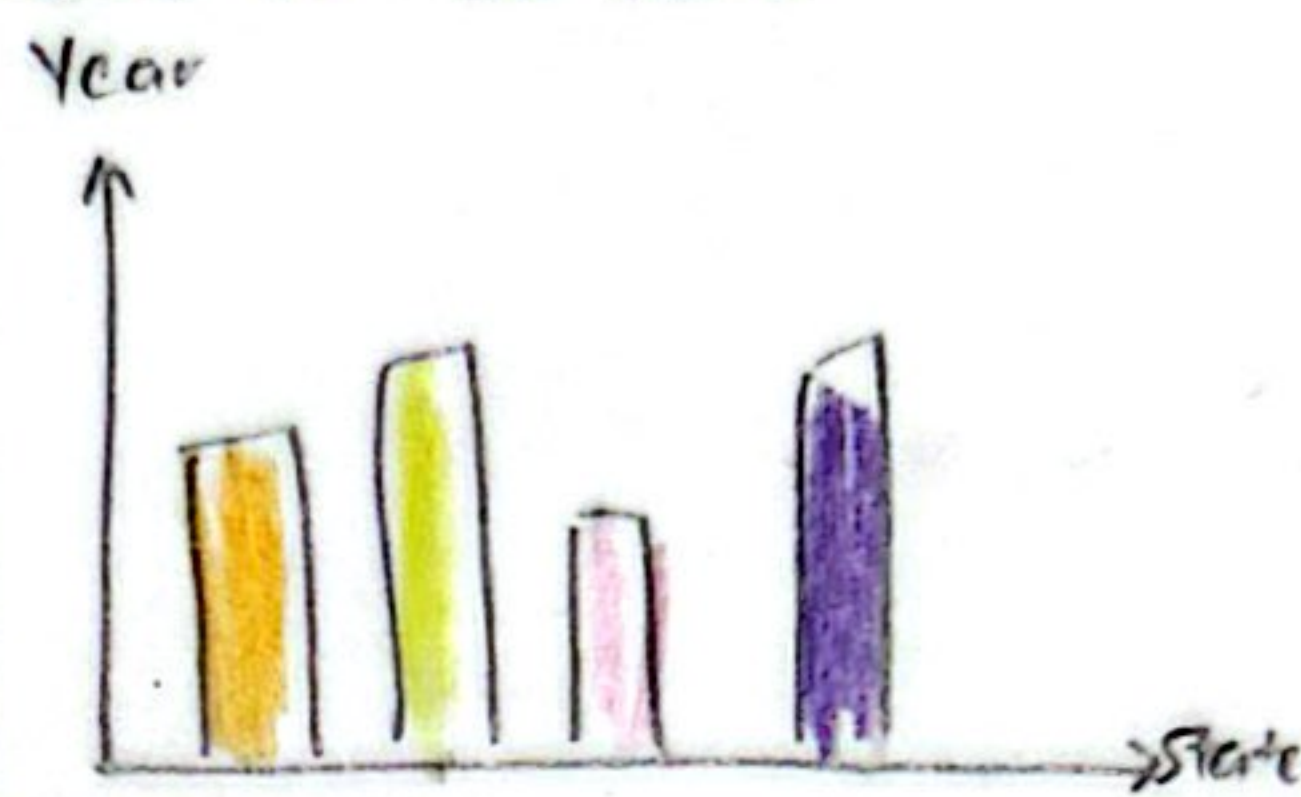
Total GHG emission by state



Temporal trend for state emission



% change pre/during/post covid for all state



Emission intensity vs GSP



Emission intensity by state & year



Focus

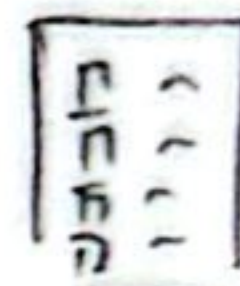
- Key techniques
 - uses linked, multi-view visualisation, where chart update dynamically thru interaction r.e. it combines geographic (map), temporal (line), comparative (bar), analytical (bubble), & pattern-based (heatmap) views.
- Novel Visualisation Aspects
 - dashboard uses coordinated filter, selecting a state on map updates line, bubble, bar, & heatmap views - This tells cohesive story of each state's emissions b4, during, aft covid
- Special detail
 - consistent clr scheme → red to green scale for emission intensity across all charts, ensuring visual continuity
 - Year slider → allow quick comparison across time for both map & heatmap
 - Dynamic annotation → highlight key moment
 - TOOLTIP harmonisation → share similar tooltip formatting
- Combination effect (provide full analytic narrative)
 - choropleth locates where emission are high
 - line shows how emission changed over time
 - bar compares overall % changes between states
 - heatmap summarises all state's yearly changes in one compact view

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Operation

- Hover on map
 - Tooltip shows state name, total emission & % change
- Click state (on map)
 - update line, bar, & heatmap for that state
- Hover on bar/bubble
 - Displays exact value & comparison to national avr
- Use year slider (2017 - 2022)

Year
- Toggle legend
 - Switch between "total emission" & "per capita emission"



Discussion

Pros

- comprehensive story combining geography, time, comparison & economics
- highly interactive & insightful for general audience & policy user
- uses consistent mapping & coordinated filtering to improve clarity

Cons

- complex layout; risk of visual clutter if spacing & clr not balanced
- requires careful performance optimisation in Vega-Lite to load smoothly

Layout

Tracing Australia's GHG story

GHG Emission
by state



GHG Total emission by state



Sector contribution over time
Emission



Emission intensity
by state



AUS - Total (net) emission
Emission



AUS - Sector emission by year
Emission



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Operation

- ① State dropdown
→ select state
(filter all chart)
- ② Year slider
→ Adjust yr
(update map)
- ③ map
→ hover
(show fig CO₂-e)
- ④ line
→ hover line
(highlight trend)
- ⑤ stacked area
→ view sector
(compare emission)
- ⑥ heatmap
→ hover tile
(show intensity)
- ⑦ stacked bar
→ hover size
(show % share)

Detail

Data: DCCBW & ABS
(2017-2022)
Via GITHUB CSVs
Tools: Vega-Lite v5,
Vega-Embed
V6, HTML/CSS

Transform: Sum, mean,
per capita,
sector share

Requirement:

- 1280 x 720 display
- GITHUB Pages hosting
- colour-blind friendly palette

Effort: ~35 hrs total

Next step:

- ① Finalise CSVs & legend
- ② Embed JSON spec
- ③ Test interactivity & UI
- ④ Deploy to Github Pages

Focus

Essence of idea: To reveal how geography, sector, & population shape Aus GHG trajectory from 2017-2022

core principles: ① exploratory storytelling
→ enabling discovery from national to state level
→ Comparative insight - linking per capita views for deeper understanding
→ clarity + accessibility - colour blind-safe palette, clear annotation, responsive layout

The dashboard "parti" (central design idea) is to visualise emission decline & efficiency improvement as a journey, emphasising both scale (total emission) & fairness (per-person impact)